NemoFleet · End-to-End NVIDIA Stack System Architecture

NVIDIA CUDA-X (nvImageCodec + CV-CUDA + TensorRT) · Cosmos-Predict2.5 / Cosmos3 VLM · Nemotron 3 (Nano 30B · Super 120B · Ultra 253B) · PhysicalAI-SmartSpaces · Nebius H200

NEBIUS × NVIDIA

16:9 SYSTEM ARCHITECTURE

LAYER 1 · OPERATOR CONSOLE & CLOUDFLARE EDGE STATE (nemofleet.pages.dev · Workers API)

CONSOLE UI
Spatial Command Map & HUD
Live floor canvas · robot halt/quarantine
4-cam ego feeds · scenario runner

GOVERNANCE
Autonomy & Adapter Controls
Measured P(actions) >= τ gate selector
Promote / Hold H200 LoRA adapters

HUMAN-IN-LOOP
Incident & Override Inspector
Accept / Override / Release alarms
Operator override writes precedent

SQL STORE
Cloudflare D1 (MEMORY)
decisions · calls · alarms · rules
1-call-per-cell precedent lookup

KV STORE
Cloudflare KV (TAGS)
Active quarantine zones & TTLs
Sub-ms edge broadcast to fleet

OBJECT STORE
Cloudflare R2 (RUNS)
Scenario replay traces & logs
Night Shift audit artifacts

store cell decision (!person)

LAYER 2 · EDGE FLEET & STAGE 1 NVIDIA CUDA-X VISION PRE-GATE (0.80 ms on H200)

Autonomous Robot Fleet
Warehouse · Data Center · SF
R14 Hazard · R03 Person · R12 Peer · R07 Clear
POST /api/events (Bearer)
↔ Countercurrent pass: 0 calls

JPEG

I/O STREAM
1080p Ego-Camera Stream
RGB JPEG frame + cell + pose
tools/ego_mix/cuda_x_pipeline.py

raw bytes

CUDA-X · CV-CUDA
NVIDIA nvImageCodec + CV-CUDA
GPU JPEG decode → NHWC tensor
cvcuda.customcrop ROI [18..82%, 45..92%]
cvcuda.gaussian(5x5) + normalize · 0.38 ms

DLPack (0-copy)

CUDA-X · TENSORRT
NVIDIA TensorRT (torch_tensorrt)
Zero-copy DLPack → PyTorch CUDA
CorridorAnomalyHead FP16/BF16
floor_anomaly_score · 0.42 ms

score >= -0.75 (wake VLM)

Gate 1 · CUDA-X ROI
anomaly_score >= -0.75?
12/12 frames · 0.80 ms

< -0.75

SCORE < -0.75
Clear Corridor Fast-Pass
Skip Cosmos VLM & Ultra (25%)
Continue patrol · 0 VLM tokens

pass-by

PEER MEMORY
System 1.5 · Peer Handshake
Antennation swap on aisle pass
Outbound gets hazards ahead
Inbound gets policy & map tags
0 LLM calls · bounded packet

peer-shared cell decisions (0 model calls)

LAYER 3 · STAGE 2 NVIDIA COSMOS PERCEPTION & NEMOTRON TIERED COGNITIVE RUNTIME

COSMOS VLM
NVIDIA Cosmos-Predict2.5 / Cosmos3
Vision-Language Scene Reasoner VLM
Extracts: person, hazard_kind, severity
Woken ONLY when CUDA-X score >= -0.75

hazard · person JSON

EVENT BUS
Structured Incident & Track Context
Scene graph + ego/overhead track kin.
functions/api/[[path]].js · runEvent()

Gate 2 · Fleet Memory
Live D1 decision at cell
& !person in scene?

cell precedent

MISS

SYSTEM 1 · REFLEX
NVIDIA Nemotron 3 Nano 30B-A3B-BF16
Hybrid Mamba-2 + Transformer MoE (3.5B active)
Dual LoRA on H200 (tools/ego_mix/serve.py):
• decide-1: JSON actions + P(actions) logprob
• plan-1 (inway): logit(true)-logit(false) >= -0.75

actions + P(actions)

Gate 3 · Safety & Gate τ
person || in_way >= -0.75
|| P(actions) < τ (0.80)?

PASS (>= τ)

ESCALATE (person || P < τ)

HIT: reuse stored cell decision (0 model calls)

quarantine tag

SYSTEM 2 · DELIBERATOR
NVIDIA Nemotron 3 Ultra 253B-A22B
Deep Deliberator (1,000-tok reasoning budget)
Handles human-in-aisle safety & novel hazards
Writes reusable cell decision to D1 Fleet Memory
Never caches person events (always re-verifies)

deliberated actions

EXECUTOR
Validated Fleet Action Executor
halt · reroute · quarantine_zone · broadcast_peer
Writes KV zone tags + D1 call & action audit log

goals + constraints

MOTION PLANNER
A* + Conflict-Based Search (CBS)
Deterministic multi-agent pathfinding around
quarantined cells & countercurrent aisle pairing

collision-free trajectories & quarantine avoidance

Promote LoRA (decide-1 + plan-1)

LAYER 4 · NIGHT SHIFT SELF-IMPROVEMENT LOOP & NEBIUS H200 SXM (141 GB) INFRASTRUCTURE

PHYSICAL AI CORPUS
NVIDIA-PhysicalAI-SmartSpaces
+ JRDB + Thör-Magni + Waymo Open
1,594,815 real trajectories
1,458 warehouse/campus 10s windows
Held-out place split (train/val/test)

tracks

REPLAY
1 · Night Shift Replay Audit
Ultra 253B re-grades today's Nano
decisions from R2 / D1 call logs
Flags disagreements for SFT
tools/night_shift/distill.cjs

hard cases

DREAM GENERATOR
NVIDIA Nemotron 3 Super 120B
2 · Counterfactual Dream Synthesis
2,048-tok reasoning generates
adversarial multi-hazard drills
Solved by Ultra → training pairs

3 · Thymus Gate
Valid schema &
!unsafe / !dup?

verified

NEBIUS H200 TRAINING
NeMo / HF TRL LoRA SFT on H200
1x Nebius H200 SXM (141 GB BF16)
LoRA r=16 α=32 on Mamba-2 + Attn:
• decide-1: 35.1% → 98.5% exact (+63.4 pt)
• plan-1: 27.6% → 85.7% exact · 76.9% recall

ARCHITECTURAL NOTATION

- NVIDIA GPU / Model Engine
- Routing / Safety Gate
- Database / Training Corpus
- I/O Stream / Peer Packet
- Sync Runtime Flow
- Async / Distillation Loop